

Question

Known:

1. The triple point of water is 273.16 K, and the freezing point is 273.15 K.
2. The relationship between the freezing point and pressure is $\Delta T/\Delta P = -7.432 \times 10^{-8} \text{ K/Pa}$.
3. At the triple point, the vapor pressure of water is 610.48 Pa.
4. The volume fractions of nitrogen and oxygen in the air are 79% and 21%, and their Henry's constants are $1.035 \times 10^{-5} \text{ mol} \cdot \text{L}^{-1}/\text{kPa}$ and $2.154 \times 10^{-5} \text{ mol} \cdot \text{L}^{-1}/\text{kPa}$, respectively.
5. The atmospheric pressure is 101325 Pa, and the acceleration due to gravity is taken as 9.8 m/s^2 .

Based on the known information, the problems to be calculated are:

- (a). The numerical value of the freezing point depression constant of water when all units are in SI units (keep two significant figures).
- (b). Place a cup of water saturated with air on a mountain top at an altitude of 5000 m, what is the freezing point of this cup of water (in K, keep five significant figures)?

The following options all contain two values, the first is the answer to question (a), and the second is the answer to question (b). The option with both answers correct is (calculation error less than 0.1% is considered correct):

A. All other options are incorrect

B. 1.8541, 273.13

C. 1.8541, 273.15

D. 1.8541, 273.14

E. 1.9542, 273.13

F. 1.9542, 273.14

G. 1.9542, 273.15

H. 2.0542, 273.15

I. 2.0542, 273.14

J. 2.0542, 273.13

K. 2.1542, 273.13

L. 2.1542, 273.14

M. 2.1542, 273.15

Answer

Correct Answer: **G**

Detailed Explanation

Problem (a):

The freezing point depression consists of two parts: the relationship between pressure and freezing point, and the colligative property of dilute solutions of air dissolved in water leading to freezing point depression.

CHECKPOINT

2 PTS

The freezing point depression consists of two parts: the relationship between pressure and freezing point, and the colligative property of dilute solutions of air dissolved in water leading to freezing point depression

Freezing point change caused by external pressure:

$$\Delta T_p = -7.423 \times 10^{-8} \times (101325 - 610.48) = -7.485 \times 10^{-3} K$$

CHECKPOINT

1 PTS

Freezing point change caused by external pressure $\Delta T_p = -7.485 \times 10^{-3} K$

According to Henry's law, the molar concentration c of dissolved gas is proportional to the partial pressure p of the gas

$$c = k_H \cdot p$$

CHECKPOINT

1 PTS

Henry's law $c = k_H \cdot p$

Therefore, the concentration of air in saturated water is:

$$c = c_{N_2} + c_{O_2} = (1.035 \times 10^{-5} \times 79\% + 2.154 \times 10^{-5} \times 21\%) \times 101.325 = 1.287 \times 10^{-3} \text{ mol/L}$$

CHECKPOINT

1 PTS

Concentration of air in saturated water $c = c_{N_2} + c_{O_2} = 1.287 \times 10^{-3} \text{ mol/L}$

For dilute aqueous solutions, the molality b is numerically approximately equal to the molarity c

CHECKPOINT

1 PTS

Dilute aqueous solutions, the molality b is numerically approximately equal to the molarity c

Freezing point change caused by colligative properties

$$\Delta T_c = K_f \cdot b$$

CHECKPOINT

1 PTS

Colligative properties $\Delta T_c = K_f \cdot b$

$$K_f = \Delta T_c / b = (\Delta T - \Delta T_p) / c = (0.01 - 7.485 \times 10^{-3}) K / (1.287 \times 10^{-3}) mol / L = 1.9542 K / kg \cdot mol$$

CHECKPOINT

2 PTS

$$K_f = 1.9542 K / kg \cdot mol$$

Problem (b):

First, it is necessary to calculate the change in pressure with altitude according to the Boltzmann distribution:

$$\ln(p_1/p_2) = -Mg\Delta h/RT$$

CHECKPOINT

1 PTS

Boltzmann distribution $\ln(p_1/p_2) = -Mg\Delta h/RT$

Where M represents the relative molecular mass of the gas.

The partial pressures of gases at 5000 m are calculated as:

$$p_{N_2,5000} = 46 \text{ kPa}, p_{O_2,5000} = 11 \text{ kPa}$$

(Since the gravitational acceleration is taken as 9.8, only two significant figures are retained subsequently)

CHECKPOINT

1 PTS

Partial pressures of gases at 5000 m $p_{N_2,5000} = 46 \text{ kPa}$, $p_{O_2,5000} = 11 \text{ kPa}$

At this time, the freezing point change caused by colligative properties is

$$\Delta T_c = K_f \cdot b = 2.0 \times (1.035 \times 10^{-5} \times 46 + 2.154 \times 10^{-5} \times 11 = 1.4 \times 10^{-3} \text{ K})$$

CHECKPOINT

1 PTS

Freezing point change caused by colligative properties $\Delta T_c = 1.4 \times 10^{-3} \text{ K}$

The freezing point change caused by pressure is

$$\Delta T_p = -7.423 \times 10^{-8} \times (46 + 11) \times 1000 = -4.2 \times 10^{-3} \text{ K}$$

CHECKPOINT

1 PTS

Freezing point change caused by pressure $\Delta T_p = -4.2 \times 10^{-3} \text{ K}$

Therefore, the total freezing point change is

$$\Delta T = -\Delta T_p + \Delta T_c = 5.6 \times 10^{-3} \text{ K}$$

The new freezing point is

$$T = 273.16 - 5.6 \times 10^{-3} = 273.15 \text{ K}$$

CHECKPOINT

1 PTS

New freezing point $T = 273.15\text{ K}$

Therefore, answer G is correct.